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Subject Name	Internet of Things	Subject Code	BCS701
Candidate Name	Varshitha A.S		USN: 1WT23CS053
Branch	Computer Science	Semester	7th
Topic	Lists, Tuples and Dictionaries		
Date of Submission		Date of Presentation	
Maximum Marks	25	Marks Obtained	
Candidate Signature		Signature of Staff with Date:	

Abstract	Lists, tuples, and dictionaries are important built-in data structures in Python programming that are used to store, organize, and manage data efficiently. Each data structure has its own characteristics and applications. Lists are ordered and mutable collections that allow duplicate elements and support modifications such as adding, removing, and updating items. Tuples are ordered and immutable collections that provide a reliable way to store data that should not be changed after creation. Dictionaries store data in key-value pairs, allowing efficient access, insertion, modification, and deletion of values using unique keys. These data structures simplify data handling and help developers write organized, readable, and efficient programs. This seminar focuses on understanding the features, syntax, operations, advantages, and practical applications of lists, tuples, and dictionaries in Python. It also highlights their differences and explains how selecting the appropriate data structure can improve program design, readability, and performance. Through suitable examples, the seminar provides a clear understanding of how these fundamental data structures are used in real-world Python applications.

1. Lists in Python

A list is a built-in data structure in Python used to store multiple items in a single variable. Lists are ordered, mutable (changeable), and allow duplicate values. They are created using square brackets [ ] .

A list in Python is a built-in data structure used to store multiple items in a single variable. Lists are ordered, mutable, and allow duplicate values. They can store elements of different data types, such as integers, strings, floating-point numbers, and even other lists. Lists are created using square brackets [], and each element is accessed using its index, starting from 0.

Example:

```
fruits = ["Apple", "Mango", "Banana", "Apple"]

print(fruits)
```

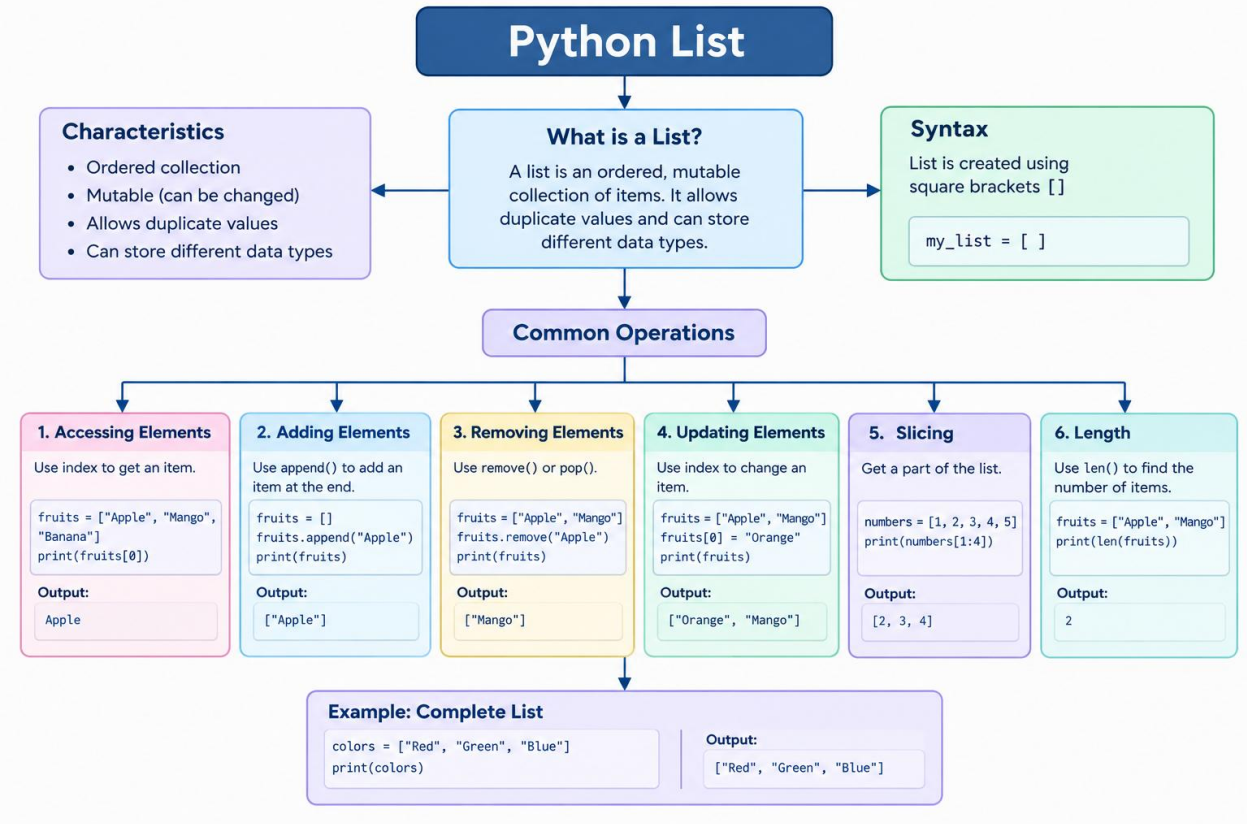
Output:

```
['Apple', 'Mango', 'Banana', 'Apple']
```

Key features:

- Ordered collection of elements.
- Allows duplicate values.
- Elements can be added, removed, or modified.
- Can store different data types.

Applications: Storing student names, marks, shopping lists, and other collections of data.



2. Tuples in Python

A tuple is a built-in data structure used to store multiple items in a single variable. Tuples are ordered and immutable (unchangeable) after creation. They are created using parentheses `()`.

Example:

```
student = ("Alice", 20, "CSE")

print(student)
```

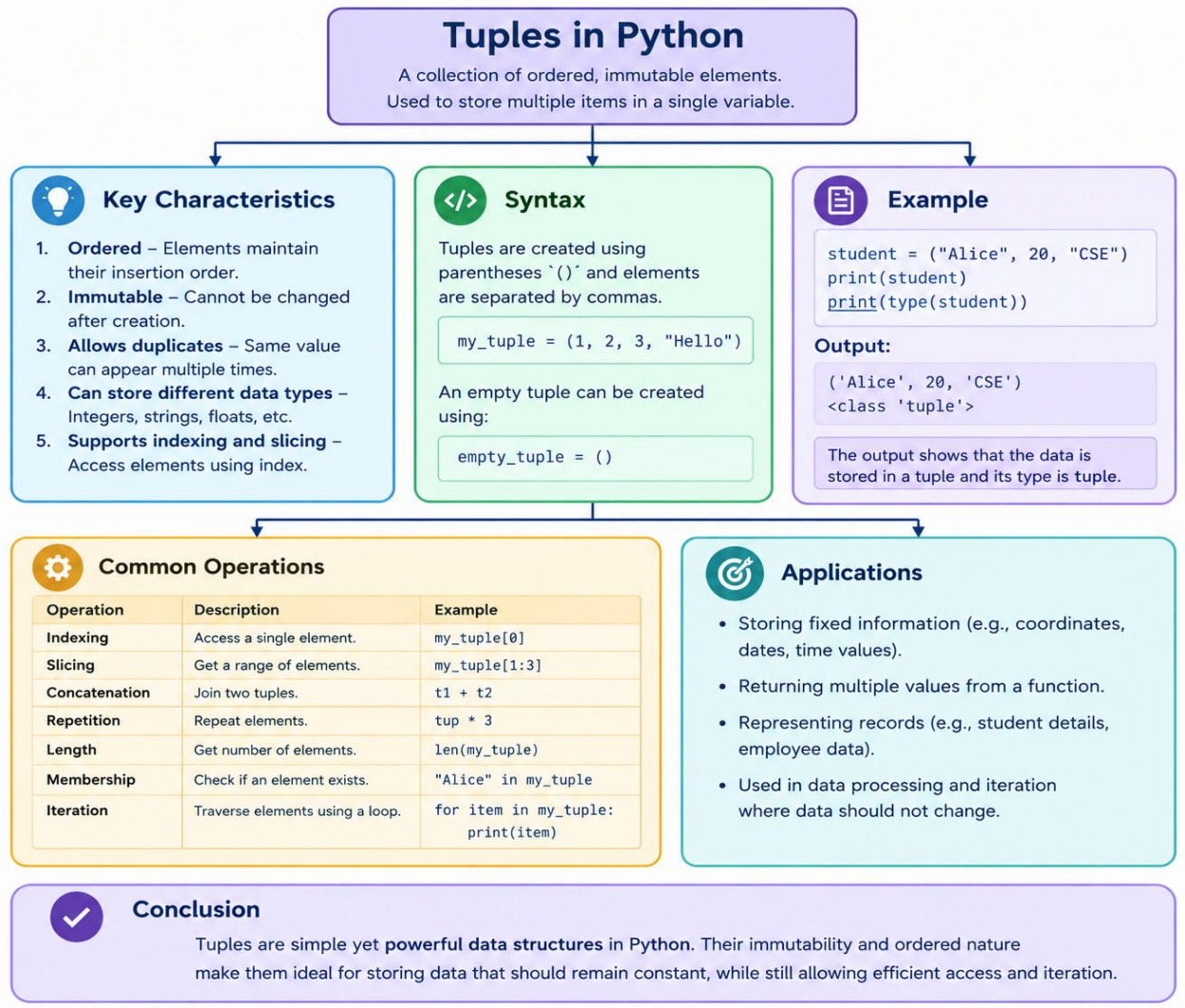
Output:

```
('Alice', 20, 'CSE')
```

Key features:

- Maintains the order of elements.
- Allows duplicate values.
- Elements cannot be changed after creation.
- Can store different data types.

Applications: Storing fixed information such as coordinates, dates, and student details that should remain unchanged.



3. Dictionaries in Python

A dictionary is a built-in data structure used to store data in key-value pairs. Each key is unique,

and values are accessed using their corresponding keys. Dictionaries are created using curly braces {}.

Example:

```
student = {
    "name": "Alice",
    "age": 20,
    "branch": "CSE"
}
print(student["name"])
```

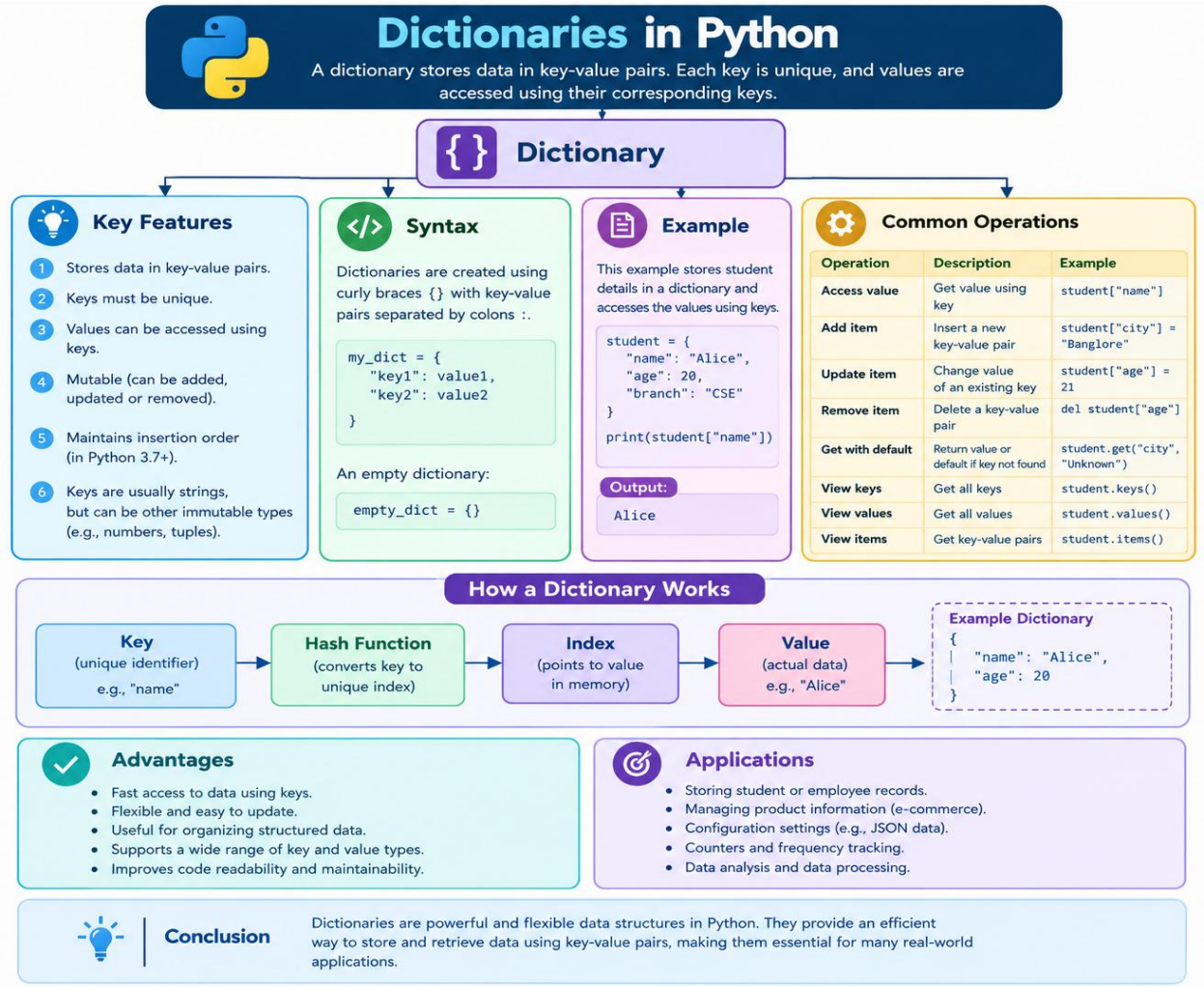
Output:

Alice

Key features:

- Stores data as key-value pairs.
- Values can be accessed using keys.
- Mutable, so items can be added, updated, or removed.
- Keys must be unique.

Applications: Storing student records, employee details, product information, and other structured data.



Conclusion

Dictionaries are an important built-in data structure in Python used to store and manage data in key-value pairs. They provide efficient access to values through unique keys and allow data to be added, modified, and deleted easily. Dictionaries improve code readability and simplify the handling of structured information. They are widely used in applications such as student record management, employee databases, web development, and data processing.